

5.4 Exploring Tables of Equivalent Ratios

Lesson Introduction

 Benchmarks	MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate. MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.		 Learning Target	I can build tables of equivalent ratios incorporating an understanding of unit rate.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.AR.3.1 Given a numerical pattern, identify and write a rule that can describe the pattern as an expression.		MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.	
 Essential Questions	- How do you use ratios to describe relationships between quantities? - How do you use unit rates to describe relationships between quantities? - How do you write a ratio as a unit rate?			
 Lesson Narrative	In this lesson, students build on their understanding of ratios and unit rates through an exploration of the number of balloons it will take to float objects of various weights. Students begin by determining the unit rate of balloons to weight in grams and then apply this rate to different weights in a table. Students deepen their understanding of the relationship between ratios and unit rates by using ratios in a table to determine the unit rate for different objects, noticing the unit rate will be the same for each, indicating that the ratios are equivalent. This lesson may also be completed using Desmos Activity Builder. Teachers can choose to facilitate the lesson through the Desmos activity or the paper-based version in the workbook if that is not an option.			
 Component Summary	5.4.1 Warm-Up (~5 min.)	Students watch a video in which a man floats using balloons and ask a question they have (<i>SE p. 188</i>)	5.4.3 Guided Practice (20-25 min.)	Use tables to explore equivalent ratios and write unit rates (<i>SE p. 190</i>)
	5.4.2 Exploration (15-20 min.)	Explore how many balloons it will take objects with different weights to float and record the data in a table (<i>SE p. 189</i>)	5.4.4 Wrap-Up (~5 min.)	Students describe how to calculate how many balloons it will take a friend to float (<i>SE p. 192</i>)
 Resources	Glossary		Additional Preparation	
	<u>Key Terms</u> : N/A <u>Additional Terms</u> : N/A		Consider making an anchor chart.	
		Materials		
		(Optional) Desmos Activity Builder - Balloon Float Video for 5.4.1 Warm-Up		

 Teacher Tips	Common Misconceptions	Things to Consider
	- In 5.4.3 Guided Practice, students may not realize that the third column asks for grams per balloon, so it should be divided by balloons or grams-to-balloon rate. - Students may forget that one quantity in a unit rate is 1 unit. - Students may forget to simplify ratios with division before multiplying to find the unknown unit. - Students may invert ratios when trying to find an equivalent rate.	- Remind students to label the units when comparing two quantities in a ratio and then to keep the unit placement the same in the second ratio. This will help avoid mistakes. - Remind students that “per balloon” means “per one balloon.” The same applies to the grams-to-pound oats example. - Remind students that fractions always express a part-to-whole comparison, but ratios can express a part-to-whole comparison or a part-to-part comparison, which can be written as a to b, $a:b$, and $\frac{a}{b}$.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Notice and Wonder	MA.K12.MTR.5.1: Patterns and Structure
	5.4.1 Warm-Up and 5.4.2 Exploration are structured with a Notice and Wonder routine. Students are asked what they notice and wonder about a video of a man using balloons to float in the air.	In 5.4.2 Exploration students look at the number of balloons needed to lift objects of different weights. Then, students use an equation to generalize the relationship between the number of balloons and the weight of the object in grams.
Lesson Notes:		

5.4.1 Warm-Up: Balloon Lift

Component Overview: Play the video of the man floating with the balloons and then have students participate in a “Notice and Wonder” routine to discuss what they noticed in the video and what they wonder. Students record their answers to question 1.



5.4.1 Warm-Up: Balloon Lift

In 2015, a Canadian man attached enough balloons to his lawn chair to float up into the sky.

1. Write a question, a wonder, or a comment about this scenario to be used during a class discussion.

Answers will vary. How many balloons are there? Is this person going to stay floating forever? How long was he in the air? How long would balloons keep me in the air?



Notice and Wonder

Observe student responses either by circulating to take note or by looking at the responses submitted via the Desmos activity.

In the whole-class discussion following the Warm-Up, highlight unique questions for the class, both mathematical and non-mathematical. This is a great time to introduce the focus of the activity, which is to answer the question: “How many balloons will it take to make the person float?”

5.4.2 Exploration: Make It Float

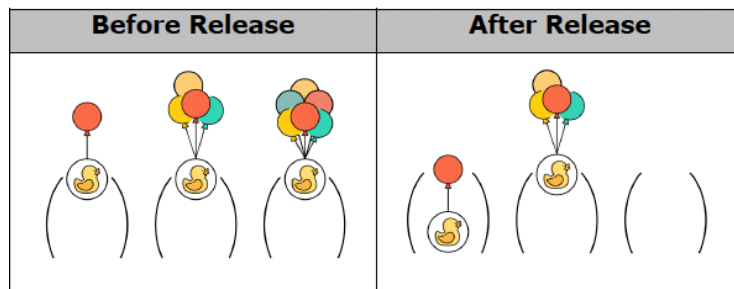
Component Overview: Students work in pairs to continue to explore the relationship between the number of balloons needed for an object to float and the object's weight in grams. Students then use two tables to make predictions about the number of balloons it will take for certain objects to float based on their weight in grams and explain their predictions. Review the second table as a class and encourage students to verify their responses and clear up any misconceptions before moving onto 5.4.3 Guided Practice.



5.4.2 Exploration: Make it Float

Flying high with balloons is dangerous! This activity will focus on determining how many balloons it takes to float various objects.

The images below show different numbers of balloons attached to rubber ducks of equal weight. The first image shows the ducks before they are released from the hold. The second image shows what happens after they are released.



1. Write one thing you know, one thing you wonder, and one thing you learned after looking at the figures above.
 - a. One thing I know is ...
Answers will vary. The duck with one balloon fell to the ground.
 - b. One thing I wonder is ...
Answers will vary. How many balloons does it take to start rising? Is it different for different weights?
 - c. One thing I learned is ...
Answers will vary. It takes 4 balloons to keep a duck floating.



To bring awareness to the importance of the ratio of balloons to weight, pose the following questions as students are working:

- What happens if there are not enough balloons?
- What happens if there are too many balloons?
- What happens if the correct number of balloons are attached to the object?

Students may not understand that the third duck floated away because it was attached to too many balloons. Be sure students understand this, so they make the connection that four is the necessary number of balloons to float.

5.4.2 Exploration: Make It Float (continued)

2. It took 4 balloons to make the duck float. The duck weighs 56 grams. If each balloon lifts the same amount, how many grams would float with 1 balloon?

14 g



3. How many balloons will it take to float a toy bear that weighs 350 grams?

a. Write your prediction in the table.

Object	Weight (grams)	Number of Balloons
Sandbag	14	1
Duck	56	4
Bear	350	25

- b. Explain the reasoning for your prediction.

Dividing the weight of the duck, 56g, by 4 balloons, the result is that 14g will float with one balloon.

With this, dividing the 350g of the bear by 14, the result is that it will take 25 balloons to make the bear float.



Think-Pair-Share

Consider structuring this component with the Think-Pair-Share routine. First, have students solve the problem individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning. Discussing the problems will help students synthesize their understanding and build student confidence.

5.4.2 Exploration: Make It Float (continued)

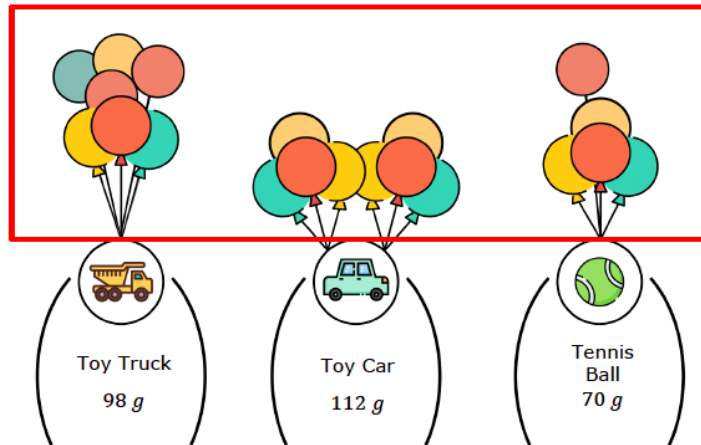
4. Complete the table for the number of balloons needed to float each object based on its weight.

Object	Weight (grams)	Number of Balloons
Sandbag	14	1
Duck	56	4
Bear	350	25
Toy Truck	98	7
Toy Car	112	8
Tennis Ball	70	5



Challenge advanced learners to find the weight of classroom objects (using a scale borrowed from the science teacher) and determine how many balloons would be needed to make it float. If a scale is not available, consider providing estimated weights for students to work with instead.

5. Draw the balloons for each object that would make it float.



5.4.3 Guided Practice: Exploring Equivalent Ratios Using Tables

Component Overview: Students work in pairs to complete the table to show the unit weight lifted by one balloon. Engage the class in a “Notice and Wonder” routine after completing the first table, then have students continue to work in pairs to determine how many balloons can lift the average adult. Students then complete another table to determine how many of a different type of balloon are needed to lift objects based on weight, using the table to determine how many of the new balloons are needed to lift an average adult.



5.4.3 Guided Practice: Exploring Equivalent Ratios Using Tables

We have looked at the number of balloons needed to lift an object of a certain weight.

1. Use the relationship between the number of balloons, B , and the weight of the object in grams, g , to write the unit rate of $\frac{\text{weight in grams}}{\text{number of balloons}}$.
 - a. Complete the table to show the unit weight lifted by a balloon.

Object	Weight (grams)	Number of Balloons	Unit Rate (grams per balloon, $\frac{g}{B}$)
Sandbag	14	1	$\frac{g}{B} = \boxed{14}$ grams per balloon
Duck	56	4	$\frac{56}{4} = 14$ grams per balloon
Bear	350	25	$\frac{350}{25} = 14$ grams per balloon
Toy Truck	98	7	$\frac{98}{7} = 14$ grams per balloon
Toy Car	112	8	$\frac{112}{8} = 14$ grams per balloon
Tennis Ball	70	5	$\frac{70}{5} = 14$ grams per balloon

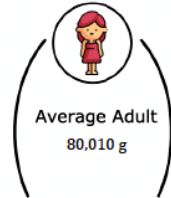


Students may struggle to set up the ratio to find the unit rate in this table because there is not a visual setup with variables, as there was in question 1. Consider guiding students with the following questions if you notice they are having trouble setting up the ratio to determine the unit rate:

- How do you determine the unit rate?
- If you are determining the weight per balloon, what does “per” mean in this instance?
- Using the answer to the previous question, where should the weight in pounds be represented in the ratio?
- Where should the number of balloons be represented in the ratio?

5.4.3 Guided Practice: Exploring Equivalent Ratios Using Tables (continued)

- b. What do you notice after completing the table?
Answers will vary. The unit rate is the same for all the objects.
2. A diagram of an average adult whose weight is 80,010 grams (g) is shown.
- a. Discuss with your partner how to use the information from the previous table to determine how many balloons it would take to float the person.
- b. Complete the statement.
 It would take **5,715** balloons to float the person.
3. Weather balloons would be better to use in this situation. Unlike regular balloons, weather balloons are designed to lift heavier objects.



The partially completed table shows the weight in pounds (lbs) of four objects and the number of weather balloons needed to float each object.

- a. Complete the table with the weight, the number of weather balloons, and unit rate for each object.

Object	Weight (lbs)	Number of Weather Balloons	Unit Rate (lbs per balloon)
Table	34	4	$\frac{34}{4} = 8.5$ lbs. per balloon
Flowerpot	17	2	$\frac{17}{2} = 8.5$ lbs. per balloon
Bookshelf	51	6	$\frac{51}{6} = 8.5$ lbs. per balloon
Armchair	93.5	11	$\frac{93.5}{11} = 8.5$ lbs. per balloon



Notice and Wonder

After students complete the table, implement a "Notice and Wonder" routine for question 1b.

- First, have students consider the table and tell a partner what they notice. Consider recording common observations on chart paper or a white board for the class to see.
- Then, have students tell their partner what the table makes them wonder.
- Share out as a class, and record on chart paper or a white board.

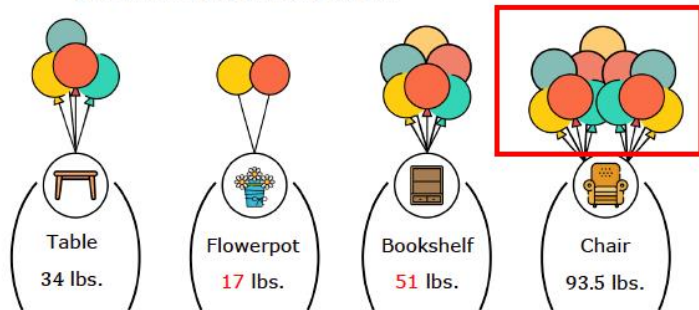
Guide students toward the conclusion that it takes one balloon to float exactly 14 grams if this has not already been discussed, and have students consider how they can use the unit rate to determine how many balloons are needed to float other objects.



The tables provide a visual entry point. Allowing students to discuss the problems provides an auditory entry point. If students have not mastered their multiplication and division facts, let them use a calculator to determine the unit rate. The calculator will allow students to focus on mastering the learning goals of this lesson.

5.4.3 Guided Practice: Exploring Equivalent Ratios Using Tables (continued)

- b. Draw the weather balloons needed to make the chair float and write the weight of each object determined from the table.



4. Consider the figure shown.



How many weather balloons would it take to float the person? Write your answer in the table.

Object	Weight (lbs)	Number of Weather Balloons	Unit Rate (lbs per balloon)
Person	204	24	$\frac{204}{24} = 8.5$ lbs. per balloon



The tables provide a visual entry point. Allowing students to discuss the problems provides an auditory entry point. If students have not mastered their multiplication and division facts, let them use a calculator to determine the unit rate. The calculator will allow students to focus on mastering the learning goals of this lesson.



Consider challenging students who may finish early to determine how many balloons it would take to float a classroom object or a pet at home.

5.4.4 Wrap-Up: Cool Down

Component Overview: Students describe how to calculate how many balloons it will take a friend to float. The Wrap-Up provides students with an opportunity to demonstrate their understanding of the learning goal of the lesson.



5.4.4 Wrap-Up: Cool Down

1. Your friend wants to know how to calculate the number of balloons she would need to float.

Answer the questions in the space provided in the table.

What information do you need to know?	<i>Her weight in pounds and the unit rate (pounds per balloon)</i>
What instructions would you give her about how to calculate the number of balloons she would need to float?	<i>To calculate the number of balloons that she would need to float, she would divide her weight by the unit rate of 8.5 lbs per balloon for weather balloons.</i>



Do you have extra time and access to tools like a scale and balloons?

Consider an extension project in which students choose classroom objects, make a conjecture for how many balloons it would take to make it float, and then mathematically prove their answers.